

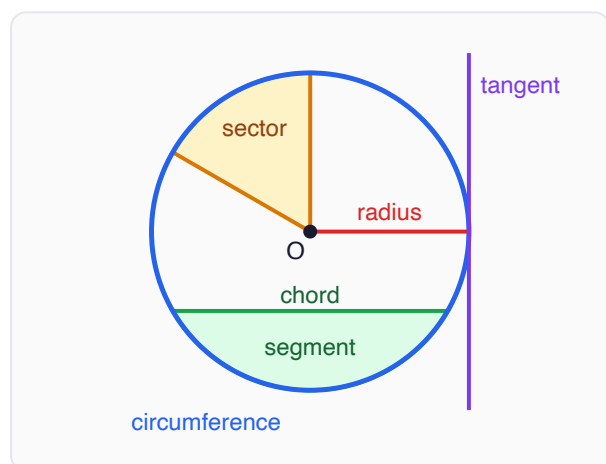
● RAG Learning

4.3 Circles & Circle Theorems

Pearson Edexcel GCSE Mathematics 1MA1 — Geometry and measures — spec refs G9, G10, G17 (circles), G18 — Papers 1 (non-calculator), 2 and 3 (calculator)

This topic covers the names of the parts of a circle (G9), the circumference and area formulae (the circle part of G17), arc lengths and sector areas (G18 — both tiers in 1MA1), and the circle theorems (G10), which are Higher-tier only. Higher students must be able to apply each theorem with a correctly worded reason and to prove the theorems.

4.3.1 Parts of a Circle (G9)



Word	Definition
Centre	The point that is the same distance from every point on the circle
Radius	A line from the centre to the circle (plural: radii)
Diameter	A chord through the centre; $d = 2r$
Circumference	The distance all the way round the circle
Chord	A straight line joining two points on the circle

Tangent	A straight line that touches the circle at exactly one point
Arc	A part of the circumference
Sector	A region enclosed by two radii and an arc
Segment	A region enclosed by a chord and an arc

Exam Tip: **Sector** = two radii + arc (a slice of pizza). **Segment** = chord + arc (the piece cut off by a straight line). Both come in *minor* (smaller) and *major* (larger) versions. A semicircle is a sector of 180° and a quadrant is a sector of 90° .

4.3.2 Circumference and Area (G17)

G17 requires you to **know** these formulae — they are not given on the 1MA1 formulae sheet.

Circumference: $C = \pi d = 2\pi r$ **Area:** $A = \pi r^2$

Rearranged: $r = \frac{C}{2\pi}$ and $r = \sqrt{\frac{A}{\pi}}$

Worked Example: A circular mirror has diameter 11 cm. Work out its circumference and its area, each correct to 1 decimal place.

$$C = \pi \times 11 = 34.55 \dots \approx \mathbf{34.6 \text{ cm}}$$

$$r = 5.5 \text{ cm, so } A = \pi \times 5.5^2 = 95.03 \dots \approx \mathbf{95.0 \text{ cm}^2}$$

Worked Example: (a) A circle has area $81\pi \text{ cm}^2$. Find its circumference in terms of π . (b) A hoop has circumference 40 cm. Find its radius to 3 significant figures.

(a) $\pi r^2 = 81\pi \Rightarrow r = 9$, so $C = 2\pi \times 9 = \mathbf{18\pi \text{ cm}}$.

(b) $r = 40 \div 2\pi = 6.366\dots \approx \mathbf{6.37 \text{ cm}}$.

Paper 1 is non-calculator, so expect "give your answer in terms of π ". Keep π as a symbol and simplify: the ring between circles of radius 9 cm and 6 cm has area $81\pi - 36\pi = 45\pi \text{ cm}^2$.

Exam Tip: Read carefully whether you are given the **radius or the diameter**. Putting the diameter into πr^2 gives an answer four times too large.

4.3.3 Arc Length and Sector Area (G18)

In 1MA1, arcs and sectors are on both tiers. A sector with angle θ is the fraction $\frac{\theta}{360}$ of the full circle, so both its arc and its area are that fraction of the whole.

$$\text{Arc length} = \frac{\theta}{360} \times 2\pi r \quad \text{Sector area} = \frac{\theta}{360} \times \pi r^2$$

$$\text{Perimeter of a sector} = \text{arc length} + r + r$$

Worked Example: A sector has radius 9 cm and angle 80° . Work out (a) its arc length, (b) its area, (c) its perimeter.

(a) $\frac{80}{360} \times 2\pi \times 9 = \frac{2}{9} \times 18\pi = 4\pi \approx \mathbf{12.6 \text{ cm}}$

(b) $\frac{80}{360} \times \pi \times 9^2 = \frac{2}{9} \times 81\pi = 18\pi \approx \mathbf{56.5 \text{ cm}^2}$

(c) $4\pi + 9 + 9 = 4\pi + 18 \approx \mathbf{30.6 \text{ cm}}$

Finding the angle or the radius

Worked Example: A sector of radius 7 cm has area 30 cm^2 . Work out the angle of the sector.

$$\frac{\theta}{360} \times \pi \times 49 = 30 \Rightarrow \theta = \frac{30 \times 360}{49\pi} = 70.15 \dots \approx \mathbf{70.2^\circ}$$

Worked Example: A sector has angle 150° and area $15\pi \text{ cm}^2$. Work out its radius.

$$\frac{150}{360} \times \pi r^2 = 15\pi \Rightarrow \frac{5}{12} r^2 = 15 \Rightarrow r^2 = 36 \Rightarrow r = \mathbf{6 \text{ cm}}$$

Segments

Area of a segment = area of the sector — area of the triangle made by the two radii and the chord.

Worked Example: A circle has radius 8 cm and chord AB subtends 90° at the centre. Work out the area of the minor segment.

$$\begin{aligned} \text{Sector} &= \frac{90}{360} \times \pi \times 64 = 16\pi. \text{ Triangle} = \frac{1}{2} \times 8 \times 8 = 32 \text{ (the radii are perpendicular).} \\ \text{Segment} &= 16\pi - 32 = 18.26 \dots \approx \mathbf{18.3 \text{ cm}^2} \end{aligned}$$

Worked Example: Higher A circle has radius 6 cm and chord AB subtends 120° at the centre. Work out the area of the minor segment. (Uses area = $\frac{1}{2}ab \sin C$, which is Higher-only.)

$$\begin{aligned} \text{Sector} &= \frac{120}{360} \times \pi \times 36 = 12\pi = 37.69 \dots \quad \text{Triangle} = \frac{1}{2} \times 6 \times 6 \times \sin 120^\circ = \\ &15.58 \dots \\ \text{Segment} &= 37.69 \dots - 15.58 \dots \approx \mathbf{22.1 \text{ cm}^2} \end{aligned}$$

Composite shapes

Worked Example: A doorway is a rectangle 0.9 m wide and 2 m high with a semicircular arch on top. Work out the perimeter and the area of the doorway, to 2 decimal places.

Arch = $\frac{1}{2} \times \pi \times 0.9 = 0.45\pi = 1.413\dots$ m. Straight edges = $0.9 + 2 + 2 = 4.9$ m (the top of the rectangle is inside the shape).

Perimeter = $4.9 + 1.413\dots \approx \mathbf{6.31\text{ m}}$.

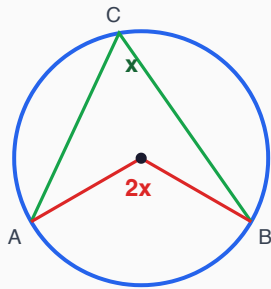
Area = $0.9 \times 2 + \frac{1}{2} \times \pi \times 0.45^2 = 1.8 + 0.318\dots \approx \mathbf{2.12\text{ m}^2}$.

Exam Tip: For a perimeter, go round the outside edge and list every piece. The two radii of a sector and the diameter of a semicircle are the edges most often forgotten; edges that are inside the shape are never counted.

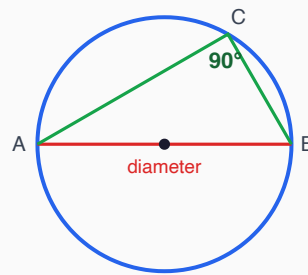
4.3.4 The Circle Theorems (G10) Higher

G10 appears only in the 1MA1 Higher content: "apply and prove the standard circle theorems concerning angles, radii, tangents and chords". When you use a theorem, write its full wording as the reason — "circle theorem" or "because of the circle" gains no credit.

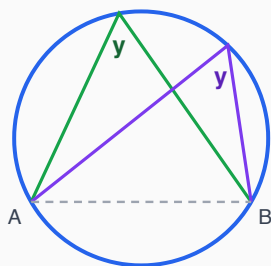
#	Theorem — the reason to write
1	The angle at the centre is twice the angle at the circumference (subtended by the same arc).
2	The angle in a semicircle is 90° .
3	Angles in the same segment are equal.
4	Opposite angles of a cyclic quadrilateral add up to 180° .
5	The tangent to a circle is perpendicular to the radius at the point of contact.
6	Tangents to a circle from an external point are equal in length.
7	The perpendicular from the centre of a circle to a chord bisects the chord.
8	The angle between a tangent and a chord is equal to the angle in the alternate segment.



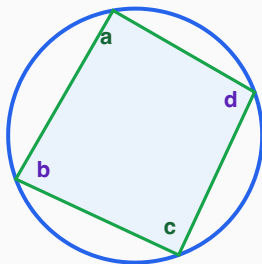
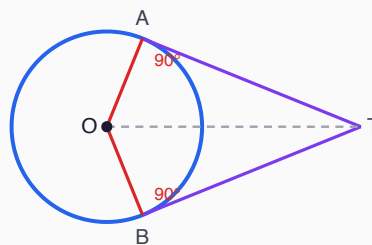
1. Centre = 2 × circumference

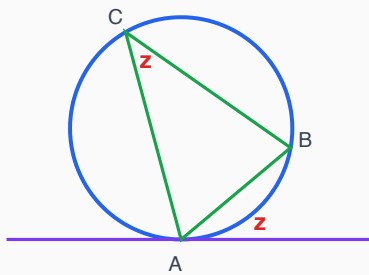


2. Angle in a semicircle



3. Angles in the same segment

4. $a + c = 180^\circ$, $b + d = 180^\circ$ 5 & 6. $OA \perp TA$ and $TA = TB$



8. Alternate segment theorem

A **cyclic quadrilateral** has all four vertices on the circumference of one circle. In theorem 8, the **alternate segment** is the segment on the opposite side of the chord from the tangent–chord angle.

Using the theorems

Worked Example: A and B lie on a circle centre O, and angle $OAB = 31^\circ$. C lies on the major arc AB. Work out angle ACB .

$OA = OB$ (radii), so triangle OAB is isosceles and angle $OBA = 31^\circ$.

Angle $AOB = 180 - 31 - 31 = 118^\circ$ (angles in a triangle add up to 180°).

Angle $ACB = 118 \div 2 = 59^\circ$ (the angle at the centre is twice the angle at the circumference).

Worked Example: TA and TB are tangents to a circle centre O at A and B. Angle $ATB = 52^\circ$. Work out angles AOB and OAB .

Angle $OAT = \text{angle } OBT = 90^\circ$ (tangent is perpendicular to the radius).

Angle $AOB = 360 - 90 - 90 - 52 = 128^\circ$ (angles in a quadrilateral add up to 360°).

$OA = OB$ (radii), so angle $OAB = (180 - 128) \div 2 = 26^\circ$.

Worked Example: ABCD is a cyclic quadrilateral. Angle $ABC = (2x + 15)^\circ$ and angle $ADC = (3x - 5)^\circ$. Work out the size of each angle.

$2x + 15 + 3x - 5 = 180$ (opposite angles of a cyclic quadrilateral add up to 180°)
 $5x + 10 = 180$, so $x = 34$. Angle $ABC = 83^\circ$, angle $ADC = 97^\circ$.

Worked Example: A chord of length 30 cm lies in a circle of radius 17 cm. How far is the chord from the centre?

The perpendicular from the centre bisects the chord, giving a right-angled triangle with hypotenuse 17 cm and one side 15 cm.

$$\text{Distance} = \sqrt{17^2 - 15^2} = \sqrt{64} = 8 \text{ cm.}$$

Worked Example: AB is a diameter of a circle and C is a point on the circumference with angle $CAB = 23^\circ$. D is another point on the circle, on the same side of chord BC as A. Work out angle ABC and angle BDC .

Angle $ACB = 90^\circ$ (angle in a semicircle), so angle $ABC = 180 - 90 - 23 = 67^\circ$ (angles in a triangle).

Angles BDC and BAC are both subtended by chord BC in the same segment, so angle $BDC = 23^\circ$ (angles in the same segment are equal).

Worked Example: The tangent at P makes an angle of 47° with the chord PQ. R lies in the alternate segment and angle $PQR = 68^\circ$. Work out angle QPR .

Angle $PRQ = 47^\circ$ (alternate segment theorem).

Angle $QPR = 180 - 47 - 68 = 65^\circ$ (angles in a triangle add up to 180°).

Exam Tip: Before calculating, mark every radius (they are all equal, so look for isosceles triangles) and every tangent–radius right angle. Write each angle on the diagram as you find it, with its reason in your working.

4.3.5 Proving the Circle Theorems (G10) Higher

Edexcel can ask you to prove a standard theorem or to use the theorems to prove a related result. Use letters for unknown angles, give a reason for every line, and end by stating exactly what was to be proved.

Worked Example: Prove that the angle at the centre is twice the angle at the circumference.

Let A, B, C be on the circle with centre O. Draw the line from C through O to a point D on the circle.

$OA = OC$ (radii), so angle $OAC = \text{angle } OCA = a$. The exterior angle of triangle OAC is angle $AOD = a + a = 2a$.

$OB = OC$ (radii), so angle $OBC = \text{angle } OCB = b$, and angle $BOD = 2b$.

So angle $AOB = 2a + 2b = 2(a + b) = 2 \times \text{angle } ACB$. ■

Worked Example: Prove that angles in the same segment are equal.

Let C and D be on the same arc, both subtending chord AB, with centre O.

Angle $ACB = \frac{1}{2}$ angle AOB and angle $ADB = \frac{1}{2}$ angle AOB (angle at the centre is twice the angle at the circumference).

Therefore angle $ACB = \text{angle } ADB$. ■

Worked Example: Prove that the opposite angles of a cyclic quadrilateral add up to 180° .

Let ABCD be cyclic with centre O; let angle $DAB = p$ and angle $BCD = q$. Draw radii OB and OD.

The angle BOD facing C is $2p$, and the reflex angle BOD facing A is $2q$ (angle at the centre is twice the angle at the circumference).

Angles around a point: $2p + 2q = 360$, so $p + q = 180^\circ$. ■

Worked Example: Prove that tangents from an external point are equal in length.

Tangents from T touch the circle, centre O, at A and B. Angle $OAT = \text{angle } OBT = 90^\circ$ (tangent is perpendicular to the radius). $OA = OB$ (radii). OT is common, and is the hypotenuse of both triangles.

So triangles OAT and OBT are congruent (RHS) and $TA = TB$. ■

Worked Example: Prove that the perpendicular from the centre to a chord bisects the chord.

Let M be the point on chord AB with $OM \perp AB$. Angle $OMA = \text{angle } OMB = 90^\circ$; $OA = OB$ (radii); OM is common.

So triangles OMA and OMB are congruent (RHS) and $AM = BM$. ■

Worked Example: Prove the alternate segment theorem.

Let the tangent at A make angle t with chord AB. Draw the diameter AE. Angle $EAT = 90^\circ$ (tangent is perpendicular to the radius), so angle $EAB = 90 - t$.

Angle $ABE = 90^\circ$ (angle in a semicircle), so angle $AEB = 180 - 90 - (90 - t) = t$.

Any angle ACB in the same segment as E equals angle AEB (angles in the same segment are equal), so angle $ACB = t$. ■

Exam Tip: A proof needs general letters, not numbers: checking one numerical case is not a proof. Proofs often rely on G5/G6 skills — isosceles triangles from radii and congruence (RHS) — so revise topic 4.2 alongside this one.

Summary

Topic	Key facts
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Parts of a circle (G9)	Radius, diameter, chord, tangent, arc, sector, segment
Circumference and area (G17)	$C = \pi d = 2\pi r, A = \pi r^2$
Arc length (G18)	$\frac{\theta}{360} \times 2\pi r$
Sector area (G18)	$\frac{\theta}{360} \times \pi r^2$; segment = sector – triangle
Circle theorems (G10) Higher	Centre = 2 × circumference; semicircle 90°; same segment equal; cyclic quadrilateral 180°; tangent ⊥ radius; equal tangents; perpendicular bisects chord; alternate segment